



Star Fox Math Mission

Grade 5 Chapter 2 Review • Multiplication

Pilot's Name: _____ Date: _____

Missions

/10



- 1 ROB's flight log is missing some factors. Fill in each missing factor.

a. $7 \times \boxed{} = 56$

b. $\boxed{} \times 9 = 63$

c. $25 \times \boxed{} = 25$

- 2 Peppy's flight rules. Name the multiplication property each one shows.

Word bank:

Commutative

Associative

Distributive

Identity

Zero

a. $38 \times 6 = 6 \times 38$

b. $243 \times 0 = 0$

c. $6 \times (40 + 3) = (6 \times 40) + (6 \times 3)$

d. $(4 \times 25) \times 9 = 4 \times (25 \times 9)$

e. $17 \times 1 = 17$

- 3 One Arwing fires $7 \times 8 = 56$ shots. Use the pattern to find each product.

a. 7×80

= $\boxed{}$

b. 7×800

= $\boxed{}$

c. $7 \times 8,000$

= $\boxed{}$

d. 50×400

= $\boxed{}$

- 4 The Great Fox has 487 fuel tanks. Each one holds 62 gallons.

- a. Round each factor, then estimate the product:

$\boxed{} \times \boxed{} = \boxed{}$

- b. Slippy says the exact answer is 3,019 gallons. Use your estimate to explain why Slippy is wrong.



- 5 Zeros in the multiplicand! Falco loads supply crates. Find each product.

a.

$$\begin{array}{r} 3,075 \\ \times 4 \\ \hline \end{array}$$

Product:

b.

$$\begin{array}{r} 4,008 \\ \times 6 \\ \hline \end{array}$$

Product:

- 6 Fox flies **348** miles on each of **27** training runs. How many miles in all?

Show your work

$$\begin{array}{r} 348 \\ \times 27 \\ \hline \end{array}$$

Miles:

- 7 **314** enemy ships each carry **526** bolts of armor. How many bolts in all?

Show your work

$$\begin{array}{r} 526 \\ \times 314 \\ \hline \end{array}$$

Bolts:

- 8 A zero in the multiplier! The **208** asteroids in the Meteo field are each **473** feet wide. Lined up end to end, how many feet long are they?

Show your work

$$\begin{array}{r} 473 \\ \times 208 \\ \hline \end{array}$$

Feet:



9

Fox buys **36** laser upgrades at **\$12.45** each. How much does he spend?

Remember the \$ and the decimal point!

Show your work

$$\begin{array}{r} \$12.45 \\ \times 36 \\ \hline \end{array}$$

Cost:



10. Final Mission • Restocking the Great Fox

Before facing Andross, Fox buys **18** Smart Bombs and **35** Shield Boosters, and pays with **\$200.00**.

How much change does he get back? Show each step.

Smart Bomb

\$4.25 each

Shield Booster

\$2.50 each

Fox pays with

\$200.00

Scratch work

1. Smart Bombs cost

2. Shield Boosters cost

3. Total Fox spends

4. Fox's change

◆ Mission complete! Andross doesn't stand a chance. ◆



1. Missing factors: Lesson 2-1, p. 66

- a. $7 \times 8 = 56$ b. $7 \times 9 = 63$ c. $25 \times 1 = 25$ (*c is the Identity Property*)

2. Name the property: Lesson 2-2, p. 68

- a. **Commutative** (*factors swapped order*) b. **Zero** (*anything times 0 is 0*)
c. **Distributive** (*the 6 multiplies both parts of $40 + 3$; both sides equal 258*)
d. **Associative** (*same order, regrouped parentheses; both sides equal 900*) e. **Identity** (*times 1 stays the same*)

3. Pattern: **560, 5,600, 56,000**, then $50 \times 400 = 20,000$ Lessons 2-3 and 2-4, pp. 70-73

Multiply the basic fact, then attach every zero from both factors. Part d is the trap: $5 \times 4 = 20$ already ends in a zero, and the three zeros from 50 and 400 still go after it. Watch for 2,000.

4. Estimating: Lesson 2-5, p. 74

- a. **$500 \times 60 = 30,000$** gallons (*487 rounds to 500, 62 rounds to 60*)
b. **3,019 is far too small.** The estimate is about 30,000, so the answer should have five digits. (*The exact answer is 30,194. Slippy probably forgot the placeholder zero when multiplying by the 6 tens of the multiplier.*)

5. Zeros in the multiplicand: **$3,075 \times 4 = 12,300$** **$4,008 \times 6 = 24,048$** Lesson 2-6, p. 76

The multiplicand is the number being multiplied (the top number). Watch for 2,448 in part b: that means a zero was skipped instead of held as a placeholder.

6. **$348 \times 27 = 9,396$** miles Lesson 2-7, p. 78

$$348 \times 7 = 2,436 \quad 348 \times 20 = 6,960 \quad 2,436 + 6,960 = 9,396$$

7. **$526 \times 314 = 165,164$** bolts Lesson 2-8, p. 80

$$526 \times 4 = 2,104 \quad 526 \times 10 = 5,260 \quad 526 \times 300 = 157,800 \quad \text{total } 165,164$$

8. **$473 \times 208 = 98,384$** feet Lesson 2-9, p. 82

$$473 \times 8 = 3,784 \quad 473 \times 200 = 94,600 \quad 3,784 + 94,600 = 98,384$$

The multiplier (208) has a 0 in the tens place. That row is all zeros and can be skipped, but the next row must still shift two places. Watch for 13,244, which comes from shifting only one place.

9. **$\$12.45 \times 36 = \448.20** Lesson 2-10, p. 84

Multiply as if it were $1245 \times 36 = 44,820$, then put back the two decimal places. Estimate check: $\$12 \times 36$ is about \$430.

10. Final Mission: Fox's change is **\$36.00** Lesson 2-11, p. 86

1. $18 \times \$4.25 = \76.50 for Smart Bombs
2. $35 \times \$2.50 = \87.50 for Shield Boosters
3. $\$76.50 + \$87.50 = \$164.00$ spent
4. $\$200.00 - \$164.00 = \$36.00$ change

The hidden step is the total cost: the problem asks for change, so the student has to find what was spent first.